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Solution

Let each side of the cube be l. The volume occupied by 3 cm depth of cube,

$$V=(3cm) \times l^2 = 3l^2 cm$$

According to the principle of floatation, we have

$$V\rho g = mg \Rightarrow V\rho = m$$

ρ is density of water = 1000 $kg\ m^{-3}$

$$\Rightarrow (3l^2 \times 10^{-2} m) \times (1000\ kg m^{-3}) = 300 \times 10^{-3} kg$$

$$l^2 = \frac{300 \times 10^{-3}}{3 \times 10^{-2} \times 1000} m^2 \Rightarrow l^2 = 100 \times 10^{-4} m^2$$

$$l = 10 \times 10^{-2} m = 10\ cm$$

Therefore, volume of cube V = l^3 = 1000 cm^3



Submarines can sink or rise in the depth of water by controlling its buoyancy. To achieve this, the submarines have ballast tanks that can be filled with water or air, alternatively, when the ballast tanks are filled with air, the overall density of the submarine becomes lesser than the surrounding water, and it surfaces (positive buoyancy). If the tanks are flooded with water replacing air, the overall density becomes greater than the surrounding water and submarine begins to sink (negative buoyancy). To keep the submarine at any depth, tanks are filled with air and water (neutral buoyancy).

Examples of floating bodies:

i) A person can swim in sea water more easily than in river water.
ii) Ice floats on water.
iii) The ship is made of steel but its interior is made hollow by giving it a concave shape.

7.4 VISCOSITY

7.4.1 Introduction

In section 7.3, the behavior of fluids at rest is discussed. Successive discussions will bring out the influence of fluid motion on different properties. A fluid in motion is a complex phenomenon, as it possesses potential, kinetic, and gravitational energy besides causing friction viscous forces to come into play. Therefore, it is necessary to consider the case of an ideal liquid to simplify the task. An ideal liquid is incompressible (i.e., bulk modulus is infinity) and in which no shearing forces can be maintained (i.e., the coefficient of viscosity is zero).

Most of the fluids offer resistance towards motion. A frictional force acts at the contact surface when a fluid moves relative to a solid or when two fluids move relative to each other. This resistance to fluid motion is similar to the friction produced when a solid moves on a surface. The internal friction existing between the layers of a moving fluid is viscosity. So, viscosity is defined as 'the property of a fluid to oppose the relative motion between its layers'.

Activity

Consider three steel balls of the same size, dropped simultaneously in three tall jars each filled with air, water, and oil respectively. It moves easily in air, but not so easily in water. Moving in oil would be even more difficult. There is a relative motion produced between the different layers of the liquid by the falling ball, which causes a viscous force. This frictional force depends